Gears: Definition, Types, Construction, Functions, and Applications

1. What Is a Gear?

A gear is a mechanical machine element with teeth on its outer or inner surface. These teeth mesh with the teeth of another gear to transmit motion and power.

In simple terms, gears transfer rotational motion and torque from one shaft to another.

Example: When a small gear turns a large gear, the large gear generally rotates more slowly and with greater torque.

2. Main Functions of Gears

3. Change speed: Increase or decrease motor speed.

4. Change torque: Reduce speed to increase output torque.

5. Change direction: Two external gears rotate in opposite directions.

6. Change the axis of rotation: Transfer motion between shafts at different angles.

7. Synchronize shafts: Maintain a fixed speed ratio between two shafts.

8. Gear Construction: Main Parts

The main parts of a typical spur gear are:

Part Function

Teeth Mesh with another gear

Gear body / web Supports the teeth

Rim Outer circular portion beneath the teeth

Hub Mounts the gear on a shaft

Bore Hole through which the shaft fits

Keyway Locks the gear to the shaft using a key

Pitch circle Reference circle used for gear ratio and meshing calculations

Root circle Circle passing through the bases of the teeth

Addendum Height from the pitch circle to the tooth tip

Dedendum Depth from the pitch circle to the tooth root

How Are Gears Made?

1. Raw material: Select steel, cast iron, bronze, or plastic.

2. Blank preparation: Make the gear’s basic cylindrical shape.

3. Tooth cutting: Cut the teeth using hobbing, shaping, or milling.

4. Heat treatment: Increase hardness and wear resistance when required.

5. Finishing: Grind and inspect the gear to achieve the required accuracy.

6. Main Types of Gears

Type 1: Spur Gear

Construction:

- The teeth are straight.

- The teeth are parallel to the gear axis.

- Spur gears operate between parallel shafts.

Function:

- Transfer rotational motion.

- Change speed and torque.

Applications:

- Simple gearboxes

- Washing machines

- Industrial machines

- Mechanical clocks

Example: A small spur gear turns a larger spur gear.

Type 2: Helical Gear

Construction:

- The teeth are cut at an angle to the gear axis.

- The teeth come into contact gradually.

- Helical gears produce axial thrust.

Function:

- Provide smooth power transmission.

- Produce less noise at high speeds.

- Allow more teeth to share the load at the same time.

Applications:

- Automobile transmissions

- Industrial gearboxes
- Compressors
- Heavy machinery

Example: Manual car transmissions commonly use helical gears.

Type 3: Bevel Gear

Construction:

- The gear is cone-shaped.
- The teeth are formed on the cone’s surface.
- Bevel gears usually transmit power between intersecting shafts.

Function:

- Change the axis of rotation.
- Often transfer power through a 90° angle.

Applications:

- Automobile differentials
- Hand drills
- Right-angle gear drives
- Machine tools

Example: Bevel gears are used in car differentials.

Type 4: Worm and Worm Wheel

Construction:

- The worm looks like a screw.
- The worm wheel is a gear.
- The two shafts are usually at 90° and do not intersect.

Function:

- Provide a large speed reduction.
- Produce low-speed, high-torque output.
- Some designs can be self-locking, but this is not guaranteed for every worm gear.

Applications:

- Lifts

- Conveyors
- Hoists
- Tuning mechanisms

Example: A worm gearbox reduces a motor’s high RPM and increases the output torque.

Type 5: Rack and Pinion

Construction:

- The pinion is a circular gear.
- The rack is a straight bar with teeth.

Function:

- Convert rotary motion into linear motion.
- Linear motion can also be converted into rotary motion.

Applications:

- Car steering systems
- CNC machines
- Linear actuators
- Sliding mechanisms

Example: Turning a car’s steering wheel moves the rack to the left or right.

Type 6: Internal Gear

Construction:

- The teeth are formed on the inside of the gear.
- A smaller external gear meshes inside it.

Function:

- Provide compact gear transmission.
- Transmit power in planetary gear mechanisms.

Applications:

- Automatic transmissions
- Planetary gearboxes
- Industrial machinery

Type 7: Planetary Gear

Construction:

A planetary gear set has four main components:

1. Sun gear: Located in the center.

2. Planet gears: Rotate around the sun gear.

3. Ring gear: The outer internal gear.

4. Planet carrier: Supports the planet gears.

Function:

- Transmit high torque in a compact space.

- Distribute the load across multiple gears.

- Provide different speed ratios.

Applications:

- Electric vehicle transmissions

- Automatic car transmissions

- Robotics

- Aerospace actuators

Type 8: Herringbone Gear

Construction:

- Has two sets of helical teeth angled in opposite directions.

- The teeth form a V shape.

Function:

- Transmit high power.

- The opposing axial thrust forces largely balance each other.

Applications:

- Heavy industrial gearboxes

- Marine propulsion

- Large compressors

5. What Is Gear Ratio?

The gear ratio tells you how the driving gear’s speed compares with the driven gear’s speed.

$$
\text{Gear Ratio}=\frac{T_2}{T_1}=\frac{N_1}{N_2}
$$

Where:

- T_1 = number of teeth on the driving gear
- T_2 = number of teeth on the driven gear
- N_1 = input speed in RPM
- N_2 = output speed in RPM

Example

Suppose:

- Small gear = 20 teeth
- Large gear = 60 teeth
- Input speed = 300 RPM

$$
GR=\frac{60}{20}=3
$$

$$
N_2=\frac{300}{3}=100\ RPM
$$